SIG
Spatial Informatics Group

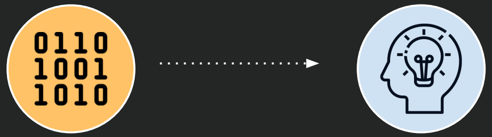
CLAY MODEL FOR DEFORESTATION DETECTION

M Warizmi Wafiq
Spatial Informatics Group

ABOUT US

What does SIG do?

We Translate Data Into Knowledge



Spatial Informatics Group (SIG) is a scientific think tank and boundary organization.

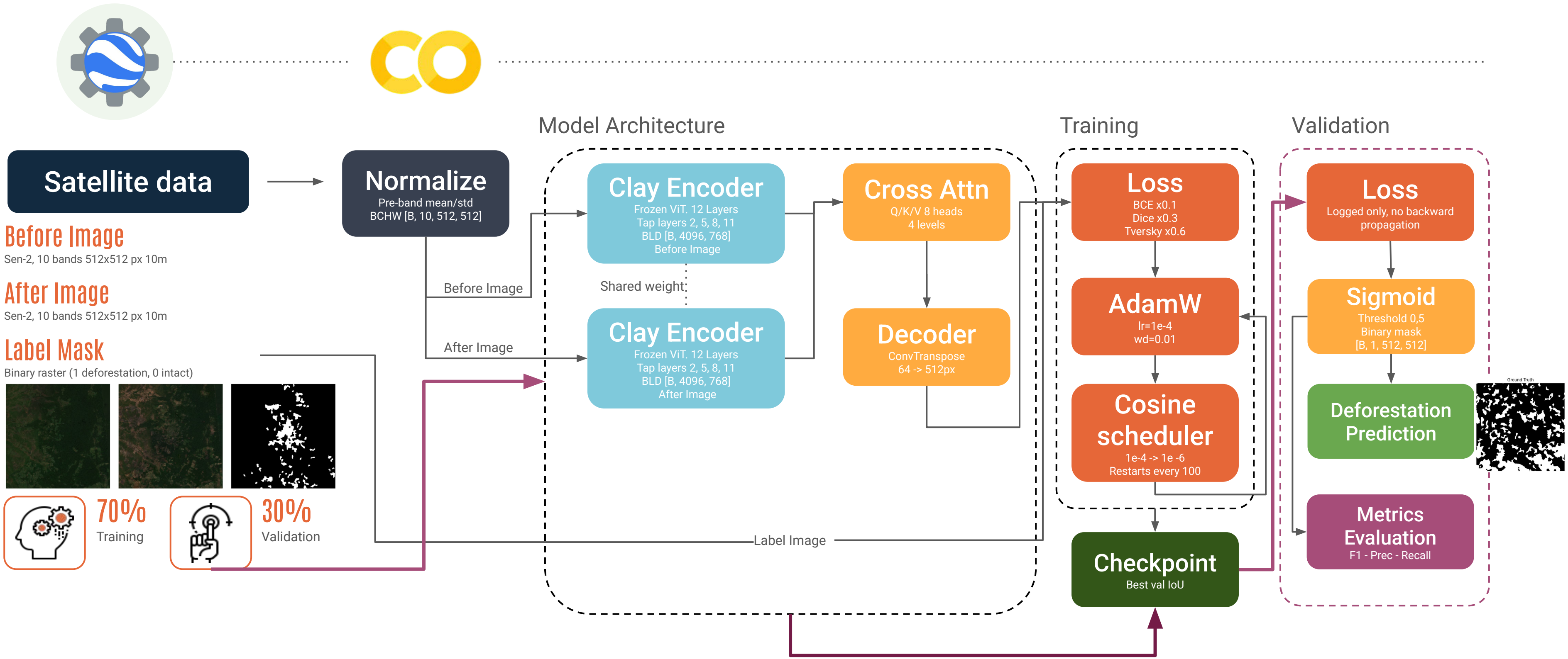
We convert geospatial data into impactful products and services, empowering decision-makers to effect positive change on a global scale. Supported by our four pillars—international talent, strategic partnerships, vertical integration, and commitment to open science—our organizational structure promotes agile collaboration among research scientists in specialized working groups.

With over 25 years in the environmental sector, we identify and address gaps in these groups through partnerships with leading institutions. Unlike traditional think-tanks, we engage upstream with universities for cutting-edge research and move downstream to produce and share products globally, often at no cost, reinforcing our dedication to knowledge dissemination and positive global impact

Clay Deforestation Model

SIG
Spatial Informatics Group

Using Machine Learning to Identify Deforestation



Clay Deforestation Model

Spatial Informatics Group

```
512×512 pixel image
        ↓
Cut into 8×8 pixel blocks

Result: 64 × 64 = 4,096 small tiles
```

```
┌─────────────────────────────────┐
│  □ □ □ □ □ □ □ □ □ □ □ □ □ □  │
│  □ □ □ □ □ □ □ □ □ □ □ □ □ □  │
│  □ □ □ □ □ □ □ □ □ □ □ □ □ □  │
│  □ □ □ □ □ □ □ □ □ □ □ □ □ □  │
│     ... 4,096 tiles total ...  │
└─────────────────────────────────┘
each □ = one 8×8 patch = 80×80 m on ground
```

```
Raw pixel values (what it IS):
  Band 1: 1200
  Band 2: 1450
  Band 3: 980
  ...

Feature vector (what it MEANS):
  Dimension 1:  0.82  ← "high probability of dense vegetation"
  Dimension 2: -0.34  ← "low urban signal"
  Dimension 3:  0.61  ← "moisture present"
  Dimension 4:  0.15  ← "texture is uniform"
  ...×768
```

```
For each patch in the After image:
  → Look at ALL patches in the Before image
  → Find which Before patches are most similar
  → Compute the difference in their feature vectors
  → High difference = something changed
  → Low difference  = no change
```

```
Patch at location [row 12, col 45]:

Before vector: [0.82, -0.34, 0.61, ...] ← dense forest signal
After  vector: [0.10, +0.72, -0.20, ...] ← bare soil signal

Difference is large → model learns: THIS IS DEFORESTATION
```

```
Patch at location [row 30, col 20]:

Before vector: [0.81, -0.33, 0.62, ...] ← dense forest
After  vector: [0.80, -0.35, 0.60, ...] ← still dense forest

Difference is tiny → model learns: NO CHANGE
```

```
4,096 patch scores (one per 80×80 m tile)
          ↓
   [Upsampling decoder — convolutional layers]
          ↓
512×512 pixel mask (one value per 10 m pixel)
```

Clay Deforestation Model

SIG
Spatial Informatics Group

```
B = Batch size        → how many images processed simultaneously
L = Length/Sequence   → number of patches (tokens) in a sequence
D = Dimension         → size of the feature vector per patch
C = Channels          → number of bands
H = Height            → number of rows (pixels)
W = Width             → number of columns (pixels)
```

```
B = 4        (batch of 4 chips)
L = 4096     (4,096 patches per image — 64×64 grid of 8×8 patches)
D = 768      (each patch encoded as a 768-dim feature vector)

Shape: [4, 4096, 768]
```

```
      D=768 columns (feature dimensions)
      ←————————————————————————————————→
   ┌──────────────────────────────────────┐   ↑
   |  patch_0:  [0.3, -0.8, 1.2, ...]    |   |
   |  patch_1:  [0.1,  0.4, -0.6, ...]   |   | L = 4096 rows
   |  patch_2:  [-0.9, 0.2, 0.7, ...]    |   | (one row per patch)
   |  ...                                |   |
   |  patch_4095: [0.5, -0.1, 0.3, ...]  |   ↓
   └──────────────────────────────────────┘
      ↑ this is ONE image (B=1)
      Repeat × 4 for the batch
```

```
B = 4      (batch of 4 chips)
C = 10     (10 Sentinel-2 bands = channels)
H = 512    (512 pixel rows)
W = 512    (512 pixel columns)

Shape: [4, 10, 512, 512]
```

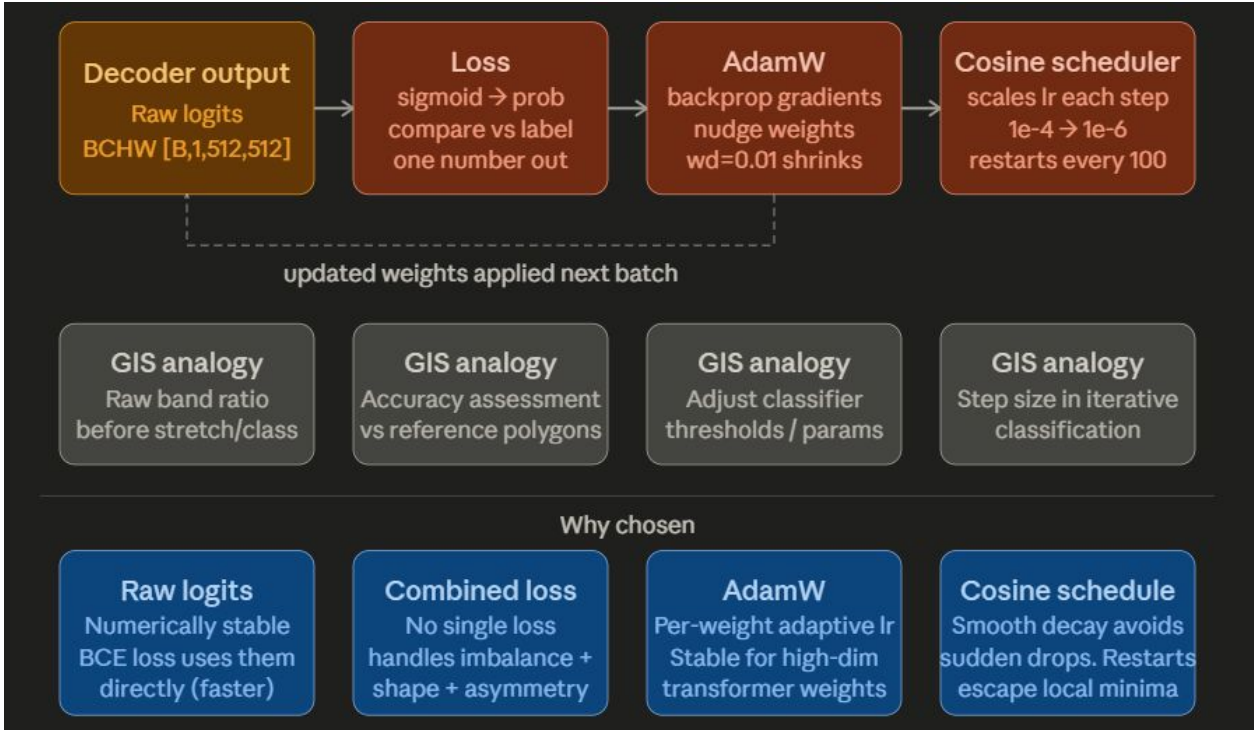
```
      W=512 columns
     ←———————————————→
  ┌─────────────────────┐   ↑
  |   Band 1 (Blue)     |   | H=512 rows
  ├─────────────────────┤   ↓
  |   Band 2 (Green)    |
  ├─────────────────────┤
  |   Band 3 (Red)      |
  ├─────────────────────┤
  |   ...               |
  ├─────────────────────┤
  |   Band 10 (SWIR2)   |
  └─────────────────────┘
 ↑ this is ONE image (B=1)
 Repeat × 4 for the batch
```

Clay Deforestation Model

Spatial Informatics Group

```
Input image:
  BCHW  [4, 10, 512, 512]
    ↓ to_patch_embed() — cut into 8×8 patches, project to 768-dim
  BLD   [4, 4096, 768]
    ↓ transformer layers (attention operates here)
  BLD   [4, 4096, 768]
    ↓ rearrange() — reshape flat list back to 2D spatial grid
  BCHW  [4, 768, 64, 64]   ← now C=768 features, H=W=64 patch grid
    ↓ output_upscaling — upsample 64→512
  BCHW  [4, 48, 512, 512]
    ↓ conv_out — classify
  BCHW  [4, 1, 512, 512]   ← final binary mask
```

Clay Deforestation Model



SIG
Spatial Informatics Group

THANK YOU!

https://sig-gis.com/